Business Problem Statement

Business scenario: Fraud & Payments Analytics at a UK Digital Bank

You're a Data/Payment Analyst at a mid-sized UK challenger bank. The Fraud & Risk Operations team and Finance team have asked you to build a reporting pack ahead of the quarterly board review, because transaction volumes have grown 40% YoY and the fraud team suspects losses are rising faster than volume.

Your brief, broken into 3 stakeholder asks:

A) Finance wants a health check on payment volumes (Q1–6, 13)
They need to know overall scale, value, split by debit/credit, and how it trends monthly — used to sanity-check revenue and liquidity forecasts.

B) Risk & Compliance want fraud exposure by channel/category (Q8–10, 20)
Following a spike in chargebacks, they want fraud rate broken down by channel (online/mobile/branch/ATM) and merchant category, to decide where to tighten controls — plus a watchlist of the riskiest transactions to review manually (ties to your AML/KYC background).

C) Growth/Regional teams want customer & regional behaviour insight (Q7, 11, 12, 14, 16, 17, 18, 19)
They want to spot high-value customers, regional performance, unusual activity (e.g. >20 transactions/month = possible money-mule pattern), and peak times — feeding into both marketing and fraud monitoring.

This maps cleanly onto your queries because it mirrors a real reporting cycle: volume/value → risk → customer/regional segmentation — exactly the kind of end-to-end deliverable a Payment/Fraud Analyst owns.

Want me to turn this into a one-page "project brief" doc (like you'd see in a real ticket), or move straight into writing the queries?

Deliverables

1. Data Preparation (Python): Clean and transform the raw dataset for analysis.

2. Data Analysis (SQL): Organize the data into a structured format, simulate business transactions, and run queries to extract insights on customer segments, loyalty, and purchase drivers.

3. Visualization & Insights (Power BI): Build an interactive dashboard that highlights key patterns and trends, enabling stakeholders to make data-driven decisions.

4. Report and Presentation: Write a clear project report summarizing your key findings and business recommendations. Prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.

5. GitHub Repository: Include all Python scripts, SQL queries, and dashboard files in a well-structured repository.